# Accelerating Azure Fleet Reliability

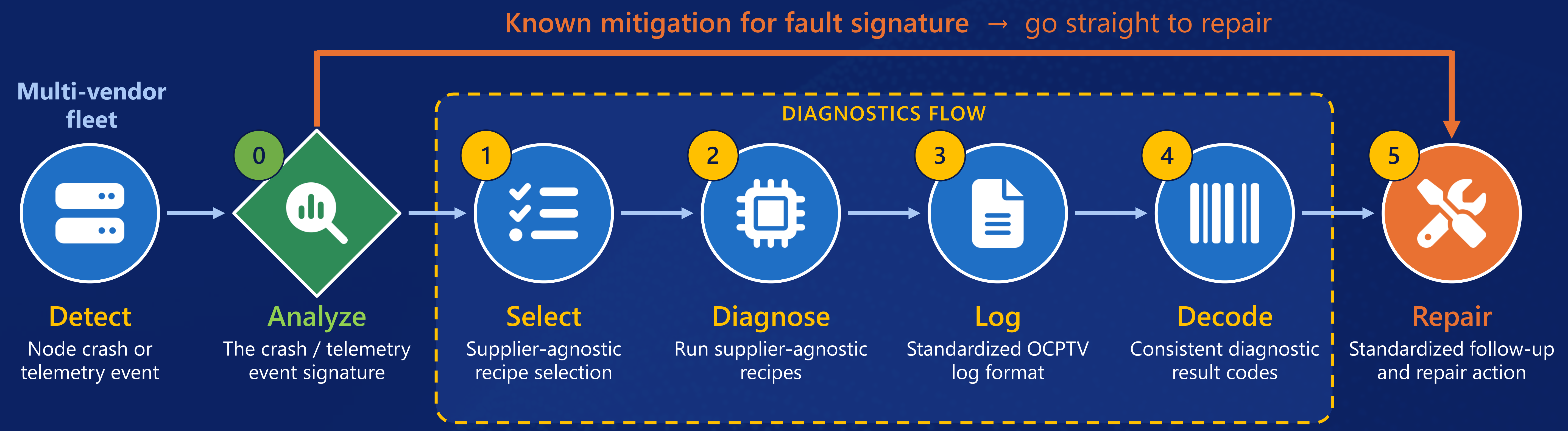
## OCP DEBUG & DIAGNOSTICS

Reducing MTTR through OCP-standardized Debug & Diagnostics innovation

Mean Time to Repair (MTTR) drives fleet availability, efficiency, and customer experience
Inconsistent, vendor-specific debug and diagnostics slow down every repair

### Reducing heterogeneous fleet complexity through standards

Diagnostics run only when there is no known repair action. Known signatures go straight to repair, reducing MTTR



Known mitigation for fault signature → go straight to repair
Multi-vendor fleet
DIAGNOSTICS FLOW
0
1
2
3
4
5

**Detect**
Node crash or telemetry event

**Analyze**
The crash / telemetry event signature

**Select**
Supplier-agnostic recipe selection

**Diagnose**
Run supplier-agnostic recipes

**Log**
Standardized OCPTV log format

**Decode**
Consistent diagnostic result codes

**Repair**
Standardized follow-up and repair action

### Supplier provided

- Fault analysis tools
- Error codes → Repair Actions through JSON
- Diag tools with recipes and OCPTV-standardized logs

### Hyperscaler managed

- **Fleet telemetry & crashdumps** from nodes
- **Analysis tools and recipe selection** based on failure
- **Repair action & FRU location** to a technician or automated agent

**Faster root cause**

**Better diagnosis**

**Heterogeneous fleet**

**Higher availability**

▲ Watch the **OCP CPACT tool demo** on the screen above

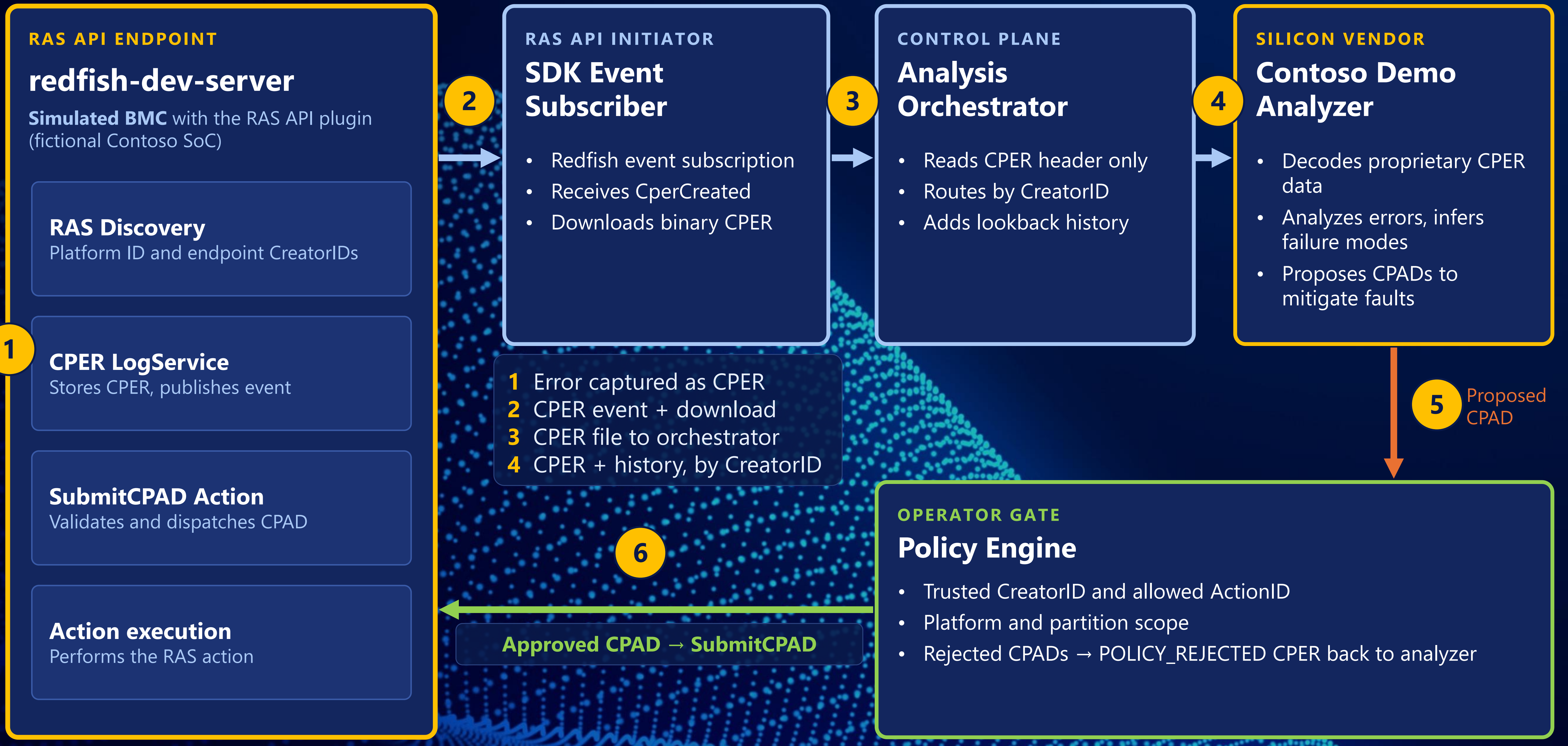
## OCP RAS API DEMO

aka.ms/ocp-ras-api-2026

OCP RAS API
RELIABILITY AT SCALE

Build your RAS pipeline once. Support any silicon.

Standard interfaces carry errors (CPERs) and actions (CPADs). The data inside can be proprietary, so silicon vendors can deliver richer logs and new RAS features



RAS API ENDPOINT
**redfish-dev-server**
**Simulated BMC** with the RAS API plugin (fictional Contoso SoC)

**RAS Discovery**
Platform ID and endpoint CreatorIDs

**CPER LogService**
Stores CPER, publishes event

**SubmitCPAD Action**
Validates and dispatches CPAD

**Action execution**
Performs the RAS action

RAS API INITIATOR
**SDK Event Subscriber**
- Redfish event subscription
- Receives CperCreated
- Downloads binary CPER

CONTROL PLANE
**Analysis Orchestrator**
- Reads CPER header only
- Routes by CreatorID
- Adds lookback history

SILICON VENDOR
**Contoso Demo Analyzer**
- Decodes proprietary CPER data
- Analyzes errors, infers failure modes
- Proposes CPADs to mitigate faults

1 Error captured as CPER
2 CPER event + download
3 CPER file to orchestrator
4 CPER + history, by CreatorID

5 Proposed CPAD

OPERATOR GATE
**Policy Engine**
- Trusted CreatorID and allowed ActionID
- Platform and partition scope
- Rejected CPADs → POLICY_REJECTED CPER back to analyzer

6 Approved CPAD → SubmitCPAD

☐ Rich data structures can contain proprietary data: Visible to endpoint and analyzer only

☐ Operator pipeline uses only the standard CPER headers and CPAD descriptors

### Silicon supplier provides

- RAS API endpoint and analyzer
- Rich error data that reflects the chip design
- CPADs for new RAS features
- Can create proprietary features

### Fleet operator builds

- One pipeline, built once against the RAS API
- Routing by standard headers and descriptors
- Policy control over every action

▲ Watch the **OCP RAS API demos** on the screen above

Redfish binding proposed • PLDM mapping in progress at DMTF • Open-source demo

Microsoft